

## EXPERIMENT-13

13. | Implement your own ping program.

**ping\_program.py**

```
import socket
import time

host = input("Enter host/IP address: ")

try:
    ip = socket.gethostbyname(host)
    print("Pinging", host, "[", ip, "]")

    for i in range(4):
        start = time.time()

        try:
            s = socket.create_connection((ip, 80), timeout=2)
        except:
            print("Request timed out.")

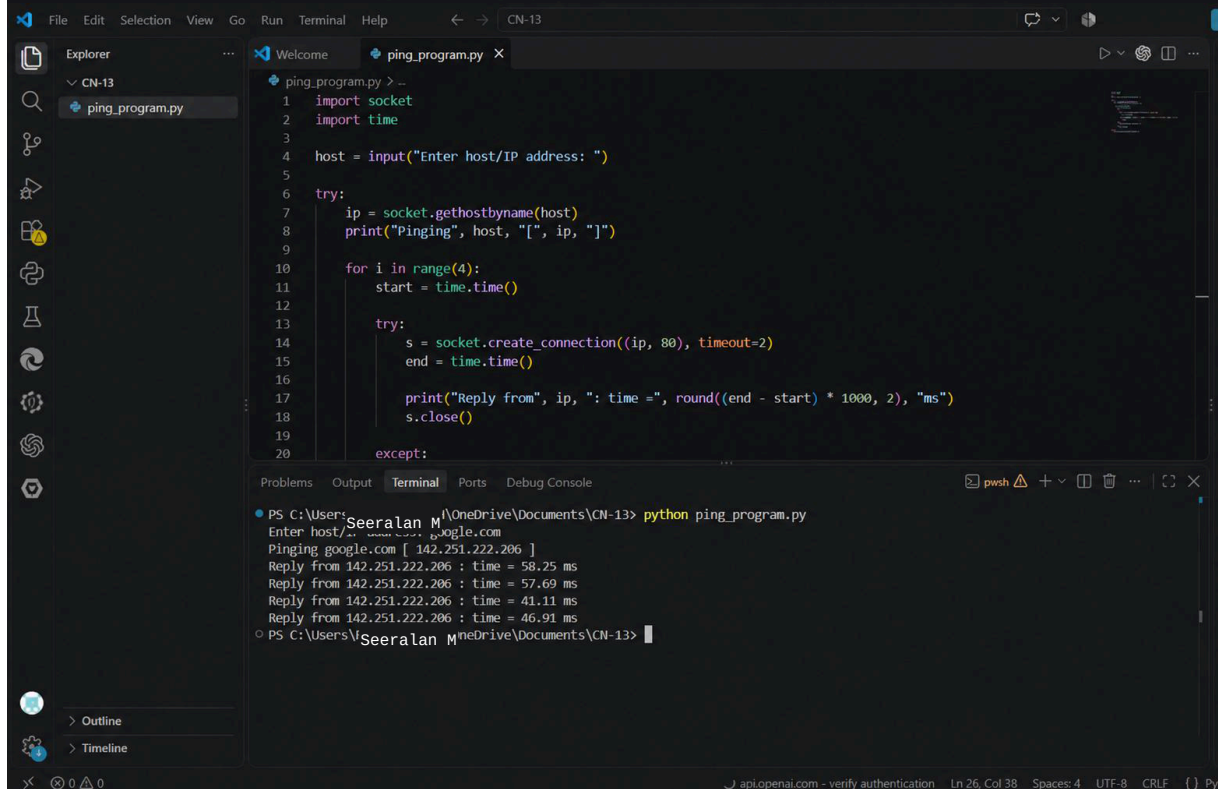
        end = time.time()
        print("Reply from", ip, ": time =", round((end - start) * 1000, 2), "ms")

        s.close()

        time.sleep(1)

except:
    print("Invalid host/IP address.")
```

```
PS C:\Users\Seeralan M\OneDrive\Documents\CN-13> python ping_program.py
Enter host/IP address: google.com
Pinging google.com [ 142.251.222.206 ]
Reply from 142.251.222.206 : time = 58.25 ms
Reply from 142.251.222.206 : time = 57.69 ms
Reply from 142.251.222.206 : time = 41.11 ms
Reply from 142.251.222.206 : time = 46.91 ms
PS C:\Users\Seeralan M\OneDrive\Documents\CN-13>
```



The screenshot displays the Visual Studio Code (VS Code) interface. The Explorer sidebar on the left shows a project named 'CN-13' with a file 'ping\_program.py'. The main editor window shows the source code of 'ping\_program.py', which is a Python script designed to ping a specified host and measure the response time. The code includes imports for 'socket' and 'time', a function to get the IP address from a host name, and a loop that performs four ping attempts, recording the time for each. The terminal at the bottom shows the command 'python ping\_program.py' being executed, followed by the program's output, which matches the text shown in the first block of the image. The status bar at the bottom indicates the file is 'api.openai.com - verify authentication' and is at line 26, column 38.

```
1 import socket
2 import time
3
4 host = input("Enter host/IP address: ")
5
6 try:
7     ip = socket.gethostbyname(host)
8     print("Pinging", host, "[", ip, "]")
9
10    for i in range(4):
11        start = time.time()
12
13        try:
14            s = socket.create_connection((ip, 80), timeout=2)
15            end = time.time()
16
17            print("Reply from", ip, ": time =", round((end - start) * 1000, 2), "ms")
18            s.close()
19
20    except:
```

```
PS C:\Users\Seeralan M\OneDrive\Documents\CN-13> python ping_program.py
Enter host/IP address: google.com
Pinging google.com [ 142.251.222.206 ]
Reply from 142.251.222.206 : time = 58.25 ms
Reply from 142.251.222.206 : time = 57.69 ms
Reply from 142.251.222.206 : time = 41.11 ms
Reply from 142.251.222.206 : time = 46.91 ms
PS C:\Users\Seeralan M\OneDrive\Documents\CN-13>
```